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https://www.tensorflow.org/api_docs/python/tf/variable_creator_scope

Scope which defines a variable creation function to be used by variable().

Function Signatures

```
tf.variable_creator_scope( variable_creator )  
def variable_creator(next_creator, **kwargs)
```

Code Examples

```
@tf_contextlib.contextmanager
tf.variable_creator_scope(
    variable_creator
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Documentation

Scope which defines a variable creation function to be used by `variable()`.

Used in the notebooks

- Validating correctness & numerical equivalence

`variable_creator` is expected to be a function with the following signature:

The creator is supposed to eventually call the `next_creator` to create a variable if it does want to create a variable and not call `Variable` or

ResourceVariable directly. This helps make creators composable. A creator may choose to create multiple variables, return already existing variables, or simply register that a variable was created and defer to the next creators in line. Creators can also modify the keyword arguments seen by the next creators.

Custom getters in the variable scope will eventually resolve down to these custom creators when they do create variables.

The valid keyword arguments in kwds are:

- `initial_value`: A Tensor, or Python object convertible to a Tensor,

which is the initial value for the Variable. The initial value must have a shape specified unless `validate_shape` is set to False. Can also be a callable with no argument that returns the initial value when called. In that case, `dtype` must be specified. (Note that initializer functions from `init_ops.py` must first be bound to a shape before being used here.)

- `trainable`: If True, the default, GradientTapes automatically watch

uses of this Variable.

- `validate_shape`: If False, allows the variable to be initialized with a

value of unknown shape. If True, the default, the shape of `initial_value` must be known.

- `caching_device`: Optional device string describing where the Variable

should be cached for reading. Defaults to the Variable's device.

If not None, caches on another device. Typical use is to cache

on the device where the Ops using the Variable reside, to deduplicate copying through Switch and other conditional statements.

- `name`: Optional name for the variable. Defaults to 'Variable' and gets

uniquified automatically.

`dtype`: If set, `initial_value` will be converted to the given type.

If `None`, either the datatype will be kept (if `initial_value` is a Tensor), or `convert_to_tensor` will decide.

- `constraint`: A constraint function to be applied to the variable after

updates by some algorithms.

- `synchronization`: Indicates when a distributed a variable will be

aggregated. Accepted values are constants defined in the class `tf.VariableSynchronization`. By default the synchronization is set to `AUTO` and the current `DistributionStrategy` chooses when to synchronize.

- `aggregation`: Indicates how a distributed variable will be aggregated.

Accepted values are constants defined in the class `tf.VariableAggregation`.

This set may grow over time, so it's important the signature of creators is as mentioned above.

Yields

